Single Point of Failure (SPOF)

find the weak link, then remove it

Watch the 3-min video

In one line: a single point of failure is any component whose loss takes the whole system down; find them by walking the request path, remove them with active-passive or active-active redundancy.

1 What a SPOF is

A chain of ten strong links breaks at its one weak link. A **SPOF** is any component that exists once and that everything depends on. Redundancy everywhere else doesn't help if one piece has no backup.

They hide in obvious places (a single database, load balancer or region) and in surprising ones: a single DNS provider, one TLS certificate about to expire, one deploy pipeline, one engineer who holds the keys.

2 Finding them

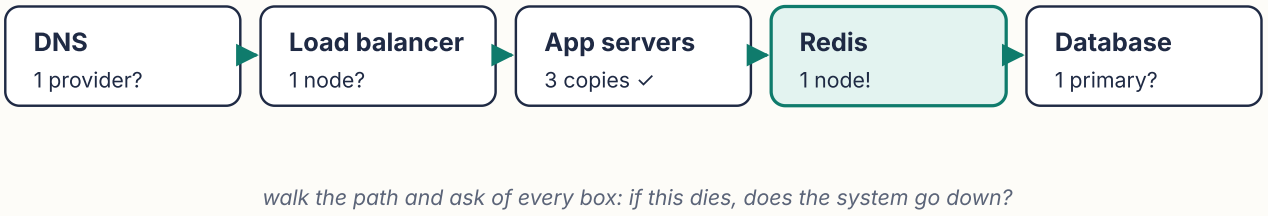


Figure 1. The app tier is redundant, but sessions live on one Redis node. When it dies, every user is logged out at once.

FIXING THE REDIS SPOF

Add a replica in another zone with **Redis Sentinel** to detect failure and promote it automatically (usually within seconds), or run **Redis Cluster** with replicas per shard. Clients must reconnect to the new primary, so use a Sentinel-aware client.

3 Two redundancy patterns

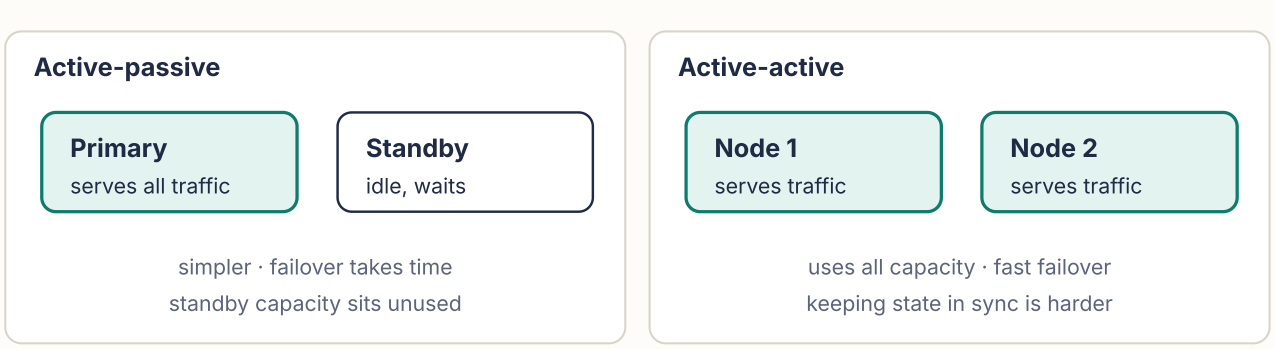


Figure 2. Active-passive is simpler; active-active uses capacity better but needs state kept in sync across nodes.

	Active-passive	Active-active
Traffic	One node at a time	All nodes at once
Failover	Seconds to minutes (detect + promote)	Near-instant: others keep serving
Cost	Standby idle	All capacity used
Hard part	Promotion without split brain	Concurrent writes and conflicts
Typical use	Primary database + standby	Stateless app servers, multi-region reads

4 The database tier

- A primary with replicas in other zones, and **automatic failover** that promotes a replica when the primary dies.
- **Backups** stored elsewhere, with tested restores.
- At larger scale, multi-region, or a distributed database designed to survive losing nodes.

COMMON MISTAKES

- Redundant app servers in front of one database or cache.
- One load balancer, or one DNS provider.
- Forgetting non-server SPOFs: certificates, secrets, CI/CD, a single person.
- A standby that was never actually failed over to in a test.

5 Key terms

SPOF: one component whose loss stops everything

Active-active: all nodes serve at once

Redis Sentinel: monitors Redis and promotes a replica

Active-passive: standby takes over on failure

Failover: switching to a backup

Split brain: two nodes both think they're primary

6 Interview questions

Q1. How do you find SPOFs? BASIC

Walk the request path (DNS, load balancer, app, cache, database) and ask of every component: if this dies, does the system go down? Include certificates, secrets, pipelines and people.

Q2. How do you remove a SPOF in the database tier? BASIC

Add replicas in other zones with automatic failover, plus tested backups; at scale, consider multi-region or a distributed database.

Q3. Active-passive vs active-active? BASIC

Active-passive: one serves, a standby takes over (simpler, slower failover, idle capacity). Active-active: all serve (better use, faster failover, harder state sync).

Q4. Every user was logged out when one Redis node restarted. Fix it. SCENARIO

Sessions had one home. Add a replica with Sentinel for automatic failover, or Redis Cluster; or use signed session tokens so the cache isn't needed to stay logged in.

Q5. Is the load balancer itself a SPOF? SCENARIO

It can be. Use a managed cloud load balancer (redundant by design) or a pair with a floating IP (VRRP/keepalived) and health checks, and spread across zones.

Q6. Only one engineer knows how to deploy. Why is that a SPOF, and what do you do? SCENARIO

If they're unavailable, nobody can ship or roll back. Automate the pipeline, document runbooks, and rotate on-call so several people can deploy.

Q7. What's split brain, and how do you prevent it? TRICKY

After a partition, two nodes both act as primary and accept conflicting writes. Prevent it with a majority quorum for promotion and fencing (the old primary is blocked from writing).

Q8. Can a region be a SPOF? TRICKY

Yes. A region-wide outage takes everything there down. Multi-region removes it, at the cost of replication latency, consistency trade-offs and higher spend; use it when the requirement justifies it.

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